Jason Cabrejos

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Experience

Burlington Northern Santa Fe – BNSF Railway *March 2023 – Present*
Robotic Embedded SW Engineer – Full Time Contractor

- Implemented a low-cost real-time inventory tracking system for under $10K per unit, merging IoT, networking, Power over Ethernet (PoE), and edge computing on the **Nvidia Jetson**
- Achieved 92% accuracy at production scale by integrating GPS, RTK, and **machine-vision/ML models** – saving $5M per year
- Scaled to multiple hardware stacks nationwide using **Docker** and **Azure IoT Hub** to manage the device fleet

Automation and Intelligent Systems Lab – UTA Research Institute *Oct 2022 – Feb 2023*
Researcher

- Developed Baxter robotic arm scripts using **ROS** and **OpenCV** for object recognition and retrieval in community-engagement projects
- Overhauled sensor stability across multiple platforms, reducing vibration-induced errors in industrial settings

Crown Castle *January 2022 – August 2022*
Robotic Development Intern

- Reduced crew size for a fiber-optic deployment operation by 30% by automating industrial equipment with **ROS**, **Python**, and Ubuntu Linux
- Bridged an industrial vehicle's CAN bus to an embedded **Nvidia Jetson** for sensor feedback and actuator control, enabling automated curb-following and cable drift correction

Laboratory of Advanced Materials Manufacturing and Analysis *June 2021 – January 2022*
Researcher

- Designed an Automated Fiber Placement (AFP) tool for a **Kuka** robotic arm for experimental composite layups
- Built slicer software to drive the AFP tool and robot arm together over **EtherNet/IP** and **Python**

Leadership, Independent Projects & Awards

SHPE Dallas/Fort Worth *June 2026 – Present*
Professional Development Director – Executive Board Officer

- Own the chapter's entire professional development pillar – strategy, budget, and delivery across three programs (PDS, TDS, GLEDA)
- Launched TDS (Technical Development Series) for 2026–2027, new certification-focused programming (PMP, PE, cloud, Six Sigma); co-founded a STEM-kit venture piloting with the chapter

Independent Projects *Self-directed, ongoing*

- **Self-hosted home lab:** administer a home server running 6+ Dockerized services (reverse proxy, uptime monitoring, workflow automation) – including a production n8n automation for SHPE DFW
- **Automated trading bot:** built and operate an unattended Python trading bot on a daily systemd timer; diagnosed and fixed a silent 3-day production outage
- **Claude Voice Assistant:** designed and shipped an end-to-end phone push-to-talk interface to an LLM assistant across 9 build phases

Hackathons *Various*

- Pitched a camera system to improve child safety without distracting the driver – **1st place**
- Presented an ML model + business plan predicting railway failures from big data – **1st place**
- Trained a mock-interviewer **LLM** with Microsoft Copilot Studio + Power Automate for practice and feedback – **2nd place**
- Trained a TensorFlow CV model to automate waste recognition/sorting on a robot – **3rd place**
- Built an agricultural IoT platform (CV + Azure + Raspberry Pi network) to schedule watering/harvest – **3rd place**

UTA Rover Team *August 2017 – August 2021*
President, Arm Leader, Member

- Led a team of 30 to the University Rover Challenge – placed 15th of 88 teams
- Managed a 5-person team through design, manufacturing, and assembly of a 5-DOF robotic arm (SolidWorks + Ansys FEA, factor of safety 2.4)
- Developed a PID joint-space controller in AutoLev, simulated in MATLAB, run on Arduino – accuracy of ±0.4 in

NASA L'Space Academy *May 2019 – November 2019*
Project Manager

- Managed two mission-concept projects: a Titan-traversing submarine and an ISS Canadarm/Dextre robotics upgrade
- Coordinated 6 students nationwide across deadlines, technical reviews, and budgeting in a dynamic engineering setting

Education

University of Texas at Arlington *August 2025*

- Master of Science in Computer Science – Intelligent Systems and Computer Networks
- Bachelor of Science in Mechanical Engineering, Minor in Computer Science

Qualifications

Programming: Python, C++, MATLAB, ROS, C#, LabVIEW
Tools & Platforms: Git, Docker, RabbitMQ, TensorFlow, Roboflow, MongoDB, Azure, AWS, IoT
AI/LLM Tooling: Claude & Copilot Studio integration, agent/automation workflows (n8n), LLM-assisted development
Modeling: SolidWorks (Certified Associate – CSWA, 2016), Solid Edge